

# Artificial Intelligence in the Knowledge Economy

Autonomy, capability, and a Lean verification exercise

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[github.com/gsaco/ai-05-ide](https://github.com/gsaco/ai-05-ide)

# Question and source

Who gains when knowledge becomes scalable?

We track bottom wages, top wages, and total output.

- Course reading: May 20, 2025, 39 pages.
- Lean source: arXiv v11, February 25, 2025, 35 pages.
- Focus: Propositions 5 and 6, then a three-type example.

# Autonomy and capability are different dimensions

## Autonomy: permitted roles

Autonomous AI can produce, work, and advise.

Co-pilot-only AI can advise.

## Capability: knowledge

$$a = z_{AI} \in [0, 1).$$

The fraction of problems AI can solve.

## Why the distinction matters

Autonomy changes outside options; capability changes which humans AI complements or substitutes for.

# The agent's problem: organize knowledge and time

A worker with knowledge  $z$  asks a solver  $s > z$  for help.

$$h n(z)(1 - z) = 1 \implies n(z) = \frac{1}{h(1 - z)}.$$

Firm profit

$$\Pi_{HH} = n(z)[s - w(z)] - w(s).$$

Humans maximize income; firms choose teams to maximize profit.

Free entry and market clearing determine wages and AI rents.

More knowledgeable workers need less help, expanding the solver's span.

# Conditions: people, problems, and firms

- Unit human mass; one unit of time per human.
- Observable  $z \in [0, 1]$  with continuous, positive density.
- Independent uniform problem difficulty; risk neutrality.
- Competition, complete information, output price one.
- No fixed costs; at most two organizational layers.

# Conditions: scarce time and abundant compute

- $0 < h < h_0$ : no independent humans before AI.
- $a < 1$ : AI is not perfectly knowledgeable.
- Production opportunities exceed production capacity.
- Compare autonomy at the same capability  $a$  and compute  $\mu$ .

A sufficient compute-abundance condition

$$\mu > \int_0^a h(1-z) dG(z) + \frac{1 - G(a)}{h(1-a)}.$$

## Proposition 5: capability determines bottom gains

Let  $B$  and  $T$  be strict wage winners below and above AI.

Bottom: a capability threshold

$$B \neq \emptyset \iff a > \bar{a}, \quad \bar{a} \in \text{int } W.$$

$W$  is the pre-AI worker set.

- Top:  $T \neq \emptyset$  for every  $a < 1$ , under the stated conditions.
- Winners occupy endpoint intervals; the human with  $z = a$  loses.
- Because  $\bar{a} \in \text{int } W$ , even some basic AI benefits the bottom.

## Proposition 6: adoption still requires capability

Co-pilot-only equilibrium has zero AI rent:  $r^N = 0$ .

| Capability    | Outcome                                                                          |
|---------------|----------------------------------------------------------------------------------|
| $a \leq w(0)$ | No adoption; wages and occupations unchanged.                                    |
| $a > w(0)$    | Lowest-knowledge humans adopt; strict bottom gains and some strict human losses. |

The adoption threshold  $w(0)$  differs from the P5 threshold  $\bar{a}$ .

## Proposition 6: bottom, top, and total output

- Near the bottom:  $w^N \geq \max\{w, w^A\}$ .  
Strict if  $a > w(0)$ .
- Near the top:  $w^N \leq w^A$ .  
Strict below  $z = 1$ .
- Total output:  $Y^A > Y^N$ .

The co-pilot equilibrium is unique, efficient, and maximizes labor income *within its permitted technology*.

# Why autonomy changes the distribution

For a human worker assisted by AI, zero profit implies

$$w = a - h(1 - z)r.$$

Autonomous AI:  $r^A = a$

AI can produce on its own.

Advice has an opportunity cost:

$$w = a[1 - h(1 - z)].$$

Co-pilot AI:  $r^N = 0$

Abundant compute can only advise.

The same team's wage becomes

$$w = a.$$

This compares team formulas; equilibrium can also change human roles.

# A three-type economy we can solve directly

| Type   | $z$   | Mass |
|--------|-------|------|
| Low    | 0     | .55  |
| Middle | $1/2$ | .30  |
| High   | 1     | .15  |

$$h = \frac{1}{2}, \quad \mu = 2.$$

Before AI, zero profit gives

$$u + \frac{1}{2}v = \frac{1}{2},$$

$$u + \frac{1}{2}t = 1,$$

$$v + \frac{1}{4}t = 1.$$

Subtract the first two:  $t - v = 1$ .

Hence  $(u, v, t) = (.2, .6, 1.6)$ .

This finite example changes the paper's population assumption.

# Autonomous AI: solve one equilibrium branch

For  $0 < a < 1/3$ , spare compute pins the rent at  $r^A = a$ .

| Human role                    | Wage                                                  |
|-------------------------------|-------------------------------------------------------|
| High: supervise AI workers    | $t = 2$                                               |
| Middle: supervise AI workers  | $v = (1 - 2a)/(1 - a)$                                |
| Low: work with middle solvers | $u = \frac{1}{2} - \frac{1}{2}v = \frac{a}{2(1 - a)}$ |

The .55 low workers use .275 middle solvers.

The remaining .025 middle humans supervise AI and pin  $v$ .

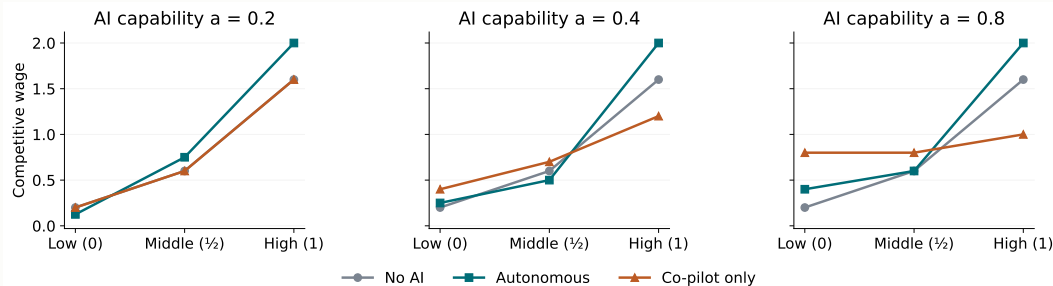
## Same autonomy, different bottom outcomes

$$\frac{a}{2(1-a)} > \frac{1}{5} \iff 5a > 2(1-a) \iff a > \frac{2}{7}.$$

| Capability | Bottom wage | Versus .20 |
|------------|-------------|------------|
| $a = .20$  | .125        | Loss       |
| $a = .30$  | 3/14        | Gain       |

The denominator is positive on  $0 < a < 1/3$ . Equality is not a gain.

# Numerical results separate the two dimensions



Across panels: change capability.

Within each panel: change autonomy.

# More output does not imply more labor income

Same capability  $a = .4$  and compute supply  $\mu = 2$ .

|               | Labor income | Compute income | Output        |
|---------------|--------------|----------------|---------------|
| No AI         | .5300        | 0              | .5300         |
| Co-pilot only | <b>.6100</b> | 0              | .6100         |
| Autonomous    | .5875        | .8000          | <b>1.3875</b> |

The numerical program checks feasibility, no profitable entry, primal–dual equality, and wage uniqueness.

# What the computational check establishes

## Allocation problem

$$\max_{x \geq 0} q^T x \quad \text{s.t.} \quad Ax \leq b.$$

Choose teams subject to human-time and compute constraints.

## Price problem

$$\min_{p \geq 0} b^T p \quad \text{s.t.} \quad A^T p \geq q.$$

Find wages and rents that rule out profitable entry.

- Equal objective values certify the finite allocation and prices.
- Wage ranges across optimal prices check uniqueness.

# The claim I would not accept without checking

## The misleading gloss

“Distribution depends on autonomy, not capability.”

$$\frac{a}{2(1-a)} > \frac{1}{5} \iff a > \frac{2}{7}, \quad 0 < a < \frac{1}{3}.$$

The genuine handwritten photograph is still pending.

**Verdict:** capability changes the sign even at fixed autonomy.

# What the paper and the example teach us

- **Bottom:** capability must cross a threshold.
- **Top:** autonomy scales expertise under the stated conditions.
- **Output:** autonomous AI permits additional productive uses.

Next: what can Lean verify?

Six slides on statements, proofs, assumptions, and remaining gaps.

# Lean 1/6: from economics to a precise statement

Paper claim  $\longrightarrow$  Lean statement  $\longrightarrow$  Proof

- A Spec : Prop states objects, assumptions, and conclusion.
- Lean checks a proof of that exact formal statement.
- Matching the statement to the paper requires a separate audit.

## Our translation boundary

Real-valued algebra and a finite wage formula are formalized.  
The continuum equilibrium and its matching structure are not.

## Lean 2/6: the bottom-gain threshold

$$0 < a < \frac{1}{3} \implies \left( \frac{1}{5} < \frac{a}{2(1-a)} \iff \frac{2}{7} < a \right).$$

```
def BottomGainSpec : Prop :=  
  ∀ a : ℝ, 0 < a → a < 1 / 3 →  
    (1 / 5 < bottomWage a ↔ 2 / 7 < a)  
  
theorem bottomGain : BottomGainSpec := by  
  intro a ha hupper  
  have hden : 0 < 2 * (1 - a) := by linarith  
  unfold bottomWage  
  rw [lt_div_iff0 hden]  
  constructor <;> intro h <;> linarith
```

`bottomWage a` means  $a/[2(1-a)]$ ; the domain is  $0 < a < 1/3$ .

**Build: PASS.** This is the finite wage inequality, not Proposition 5.

## Lean 3/6: why the inequality proof works

```
intro a ha hupper
have hden : 0 < 2 * (1 - a) := by linarith
unfold bottomWage
rw [lt_div_iff0 hden]
constructor <;> intro h <;> linarith
```

- hden: prove  $2(1 - a) > 0$  from the stated domain.
- lt\_div\_iff0: clear division without reversing the inequality.
- constructor: split “if and only if” into two directions.  
linarith closes each direction by linear arithmetic.

Identities need nonzero denominators; this inequality needs **positivity**.

## Lean 4/6: zero profit implies a wage identity

$$n(z) = \frac{1}{h(1-z)}, \quad n(z)(a-w) - r = 0 \implies w = a - h(1-z)r.$$

```
have hhne : h ≠ 0 := ne_of_gt hh
have hzne : 1 - z ≠ 0 := ne_of_gt (sub_pos.mpr hz)
unfold topAutomatedProfit span at hprofit
field_simp [hhne, hzne] at hprofit
nlinarith
```

`field_simp` uses  $h \neq 0$  and  $1 - z \neq 0$  to clear denominators.  
`nlinarith` then proves the wage identity.

**Build: PASS.** `zeroProfitWage` assumes  $0 < h < 1$ ,  $0 \leq z < 1$ , and zero profit; it does not prove an equilibrium.

## Lean 5/6: a missing assumption changes the claim

Dropping  $z < 1$  allows

$$h = \frac{1}{2}, \quad z = 1, \quad a = \frac{1}{2}, \quad w = 1, \quad r = 0.$$

- Lean's real division is total: here  $n(z) = 0$ .
- Profit is zero, but the proposed wage identity says  $1 = 1/2$ .
- The separate counterexample compiles successfully.

This tests unrestricted algebra; it is not an admissible economic team.

## Lean 6/6: what passed and what remains open

| Status  | Evidence                                             |
|---------|------------------------------------------------------|
| PASS    | Support proofs, full build, and required fast check. |
| STOPPED | Semantic preflight: missing source statement map.    |
| 0 of 6  | Full continuum propositions proved.                  |

Matching, equilibrium, sorting, and source-level audits remain open.

### What I learned

A checked proof is useful when its assumptions and scope are explicit.